- To find the *y*-intercept of the line, let $x = 0$ and solve for *y*.

x	y
	0
0	

- **Graph a line using the intercepts**
  Step 1. Find the *x*- and *y*- intercepts of the line.
    - Let $y = 0$ and solve for *x*.
    - Let $x = 0$ and solve for *y*.

  Step 2. Find a third solution to the equation.
  Step 3. Plot the three points and then check that they line up.
  Step 4. Draw the line.

- **Choose the most convenient method to graph a line**
  Step 1. Determine if the equation has only one variable. Then it is a vertical or horizontal line.
  $x = a$ is a vertical line passing through the *x*-axis at *a*.
  $y = b$ is a horizontal line passing through the *y*-axis at *b*.
  Step 2. Determine if *y* is isolated on one side of the equation. The graph by plotting points.
  Choose any three values for *x* and then solve for the corresponding *y*- values.
  Step 3. Determine if the equation is of the form $Ax + By = C$, find the intercepts.
  Find the *x*- and *y*- intercepts and then a third point.

11.4 Understand Slope of a Line

- **Find the slope from a graph**
  Step 1. Locate two points on the line whose coordinates are integers.
  Step 2. Starting with the point on the left, sketch a right triangle, with the hypotenuse going from the first point to the second point.
  Step 3. Count the rise and the run on the legs of the triangle.
  Step 4. Take the ratio of rise to run to find the slope, $m = \frac{\text{rise}}{\text{run}}$

- **Slope of a Horizontal Line**
  - The slope of a horizontal line, $y = b$, is 0.

- **Slope of a Vertical Line**
  - The slope of a vertical line, $x = a$, is undefined.

- **Slope Formula**
  - The slope of the line between two points (x_1, y_1) and (x_2, y_2) is $m = \frac{y_2 - y_1}{x_2 - x_1}$

- **Graph a line given a point and a slope.**
  Step 1. Plot the given point.
  Step 2. Use the slope formula to identify the rise and the run.
  Step 3. Starting at the given point, count out the rise and run to mark the second point.
  Step 4. Connect the points with a line.

Exercises

Review Exercises

Use the Rectangular Coordinate System

Plot Points in a Rectangular Coordinate System

In the following exercises, plot each point in a rectangular coordinate system.

277. $(1, 3), (3, 1)$

278. $(2, 5), (5, 2)$

In the following exercises, plot each point in a rectangular coordinate system and identify the quadrant in which the point is located.

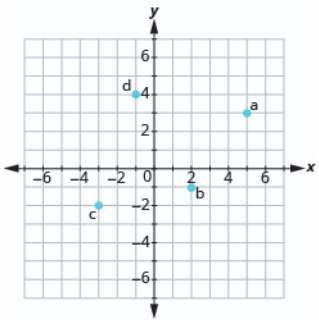
279. ⓐ $(-1,-5)$ ⓑ $(-3,4)$ ⓒ $(2,-3)$ ⓓ $\left(1,\frac{5}{2}\right)$

280. ⓐ $(3,-2)$ ⓑ $(-4,-1)$ ⓒ $(-5,4)$ ⓓ $\left(2,\frac{10}{3}\right)$

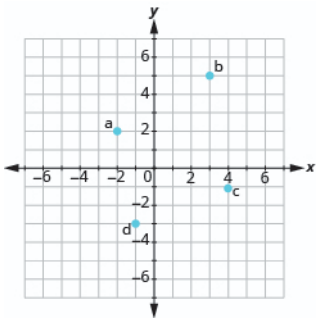
Identify Points on a Graph

In the following exercises, name the ordered pair of each point shown in the rectangular coordinate system.

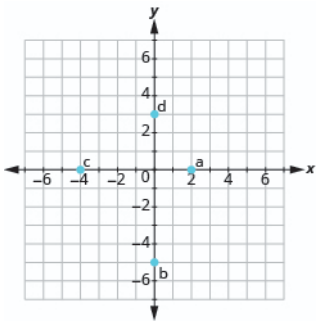
281.



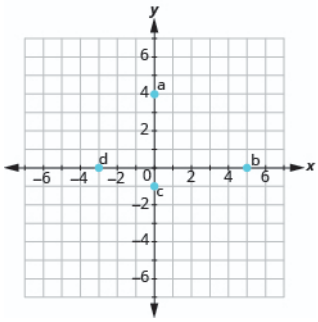
282.



283.



284.



Verify Solutions to an Equation in Two Variables

In the following exercises, find the ordered pairs that are solutions to the given equation.

285. $5x + y = 10$
ⓐ $(5,1)$ ⓑ $(2,0)$ ⓒ $(4,-10)$

286. $y = 6x - 2$
ⓐ $(1,4)$ ⓑ $\left(\frac{1}{3},0\right)$ ⓒ $(6,-2)$

Complete a Table of Solutions to a Linear Equation in Two Variables

In the following exercises, complete the table to find solutions to each linear equation.

287. $y = 4x - 1$

x	y	(x, y)
0		
1		
-2		

288. $y = -\frac{1}{2}x + 3$

x	y	(x, y)
0		
1		
-2		

289. $x + 2y = 5$

x	y	(x, y)
	0	
1		
-1		

290. $3x - 2y = 6$

x	y	(x, y)
0		
	0	
-2		

Find Solutions to a Linear Equation in Two Variables

In the following exercises, find three solutions to each linear equation.

291. $x + y = 3$ **292**. $x + y = -4$ **293**. $y = 3x + 1$

294. $y = -x - 1$

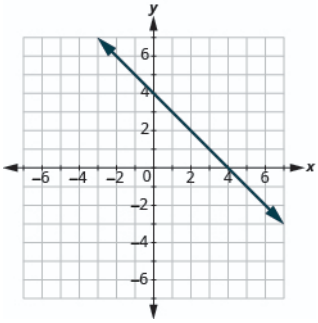
Graphing Linear Equations

Recognize the Relation Between the Solutions of an Equation and its Graph

In each of the following exercises, an equation and its graph is shown. For each ordered pair, decide

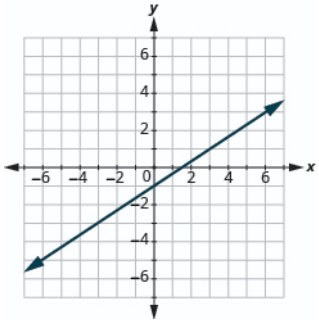
ⓐ if the ordered pair is a solution to the equation. ⓑ if the point is on the line.

295. $y = -x + 4$



1. $(0, 4)$
2. $(-1, 3)$
3. $(2, 2)$
4. $(-2, 6)$

296. $y = \frac{2}{3}x - 1$



1. $(0, -1)$
2. $(3, 1)$
3. $(-3, -3)$
4. $(6, 4)$

Graph a Linear Equation by Plotting Points

In the following exercises, graph by plotting points.

297. $y = 4x - 3$ **298**. $y = -3x$ **299**. $2x + y = 7$

Graph Vertical and Horizontal lines

In the following exercises, graph the vertical or horizontal lines.

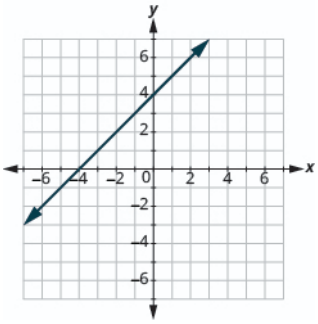
300. $y = -2$ **301**. $x = 3$

Graphing with Intercepts

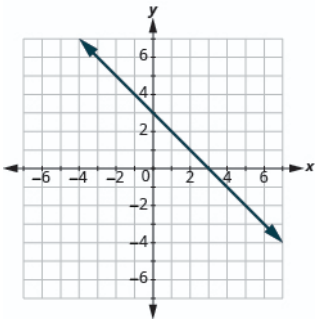
Identify the Intercepts on a Graph

In the following exercises, find the x– and y-intercepts.

302.



303.



Find the Intercepts from an Equation of a Line

In the following exercises, find the intercepts.

304. $x + y = 5$

305. $x - y = -1$

306. $y = \frac{3}{4}x - 12$

307. $y = 3x$

Graph a Line Using the Intercepts

In the following exercises, graph using the intercepts.

308. $-x + 3y = 3$

309. $x + y = -2$

Choose the Most Convenient Method to Graph a Line

In the following exercises, identify the most convenient method to graph each line.

310. $x = 5$

311. $y = -3$

312. $2x + y = 5$

313. $x - y = 2$

314. $y = \frac{1}{2}x + 2$

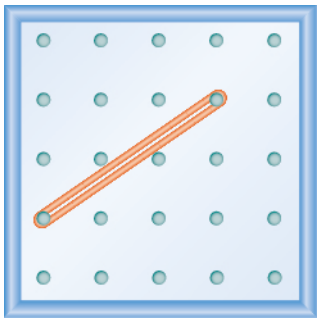
315. $y = \frac{3}{4}x - 1$

Understand Slope of a Line

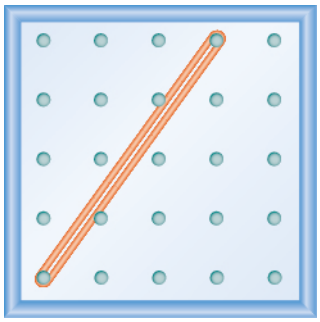
Use Geoboards to Model Slope

In the following exercises, find the slope modeled on each geoboard.

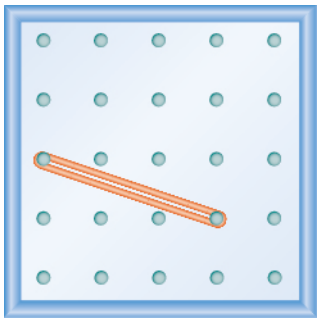
316.



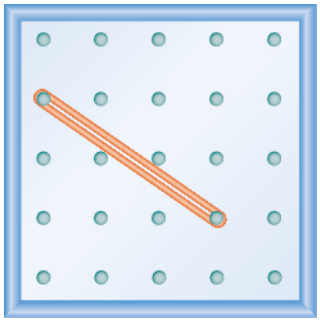
317.



318.



319.



In the following exercises, model each slope. Draw a picture to show your results.

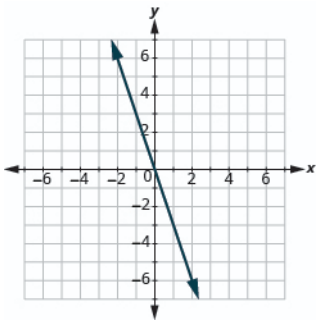
320. $\frac{1}{3}$ **321.** $\frac{3}{2}$ **322.** $-\frac{2}{3}$

323. $-\frac{1}{2}$

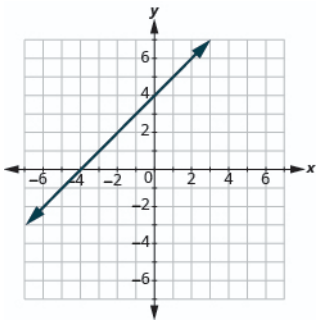
Find the Slope of a Line from its Graph

In the following exercises, find the slope of each line shown.

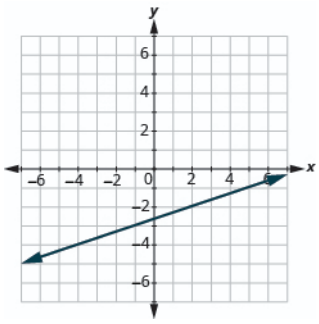
324.



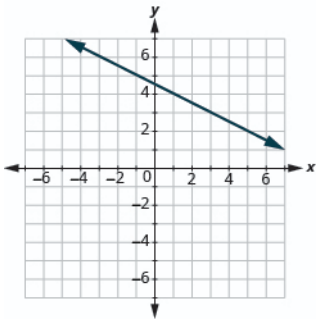
325.



326.



327.



Find the Slope of Horizontal and Vertical Lines

In the following exercises, find the slope of each line.

328. $y = 2$ **329.** $x = 5$ **330.** $x = -3$

331. $y = -1$

Use the Slope Formula to find the Slope of a Line between Two Points

In the following exercises, use the slope formula to find the slope of the line between each pair of points.

332. $(2, 1), (4, 5)$ **333.** $(-1, -1), (0, -5)$ **334.** $(3, 5), (4, -1)$

335. $(-5, -2), (3, 2)$

Graph a Line Given a Point and the Slope

In the following exercises, graph the line given a point and the slope.

336. $(2, -2); m = \frac{5}{2}$

337. $(-3, 4); m = -\frac{1}{3}$

Solve Slope Applications

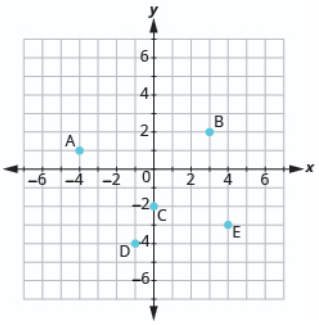
In the following exercise, solve the slope application.

338. A roof has rise 10 feet and run 15 feet. What is its slope?

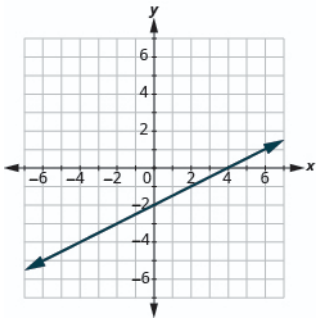
Practice Test

339. Plot and label these points:
ⓐ $(2, 5)$ ⓑ $(-1, -3)$
ⓒ $(-4, 0)$ ⓓ $(3, -5)$
ⓔ $(-2, 1)$

340. Name the ordered pair for each point shown.



341. Find the x-intercept and y-intercept on the line shown.



342. Find the x-intercept and y-intercept of the equation $3x - y = 6$.

343. Is $(1, 3)$ a solution to the equation $x + 4y = 12$? How do you know?

344. Complete the table to find four solutions to the equation $y = -x + 1$.

x	y	(x, y)
0		
1		
3		
-2		

345. Complete the table to find three solutions to the equation $4x + y = 8$

x	y	(x, y)
0		
	0	
3		

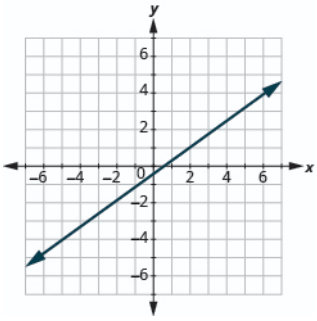
In the following exercises, find three solutions to each equation and then graph each line.

346. $y = -3x$

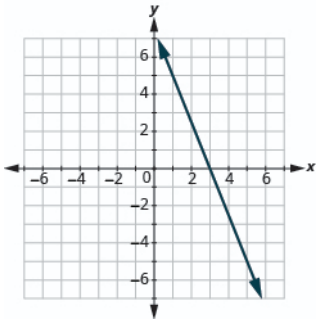
347. $2x + 3y = -6$

In the following exercises, find the slope of each line.

348.



349.



350. Use the slope formula to find the slope of the line between $(0, -4)$ and $(5, 2)$.

351. Find the slope of the line $y = 2$.

352. Graph the line passing through $(1, 1)$ with slope $m = \frac{3}{2}$.

353. A bicycle route climbs 20 feet for $1{,}000$ feet of horizontal distance. What is the slope of the route?